

AYOMIDE WILFRED ADEYEMI

Backend Software Engineer | AI Builder | Sustainability Researcher

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

RESEARCH INTERESTS

Sustainable Software Engineering • AI-Augmented Development • Human Aspects in Software Engineering • Distributed Systems Architecture • Cloud Computing & DevOps • Software Engineering for Resource-Constrained Environments

EDUCATION

Bachelor of Engineering (B.Eng.), Electrical and Electronics Engineering

2018 – 2022

Kwara State University, Nigeria | WES-Evaluated GPA: 3.58 / 4.0

Graduated with honours. Coursework in robotics, signal systems, embedded computing, and telecommunications provided a rigorous technical foundation that underpins all subsequent engineering and IT work.

PUBLISHED TECHNICAL WRITING & APPLIED RESEARCH

Peer-Reviewed Research

► [Gas Monitoring and Leakages Detection Using IoT](#)

Adeyemi, A. W., & Musa, A. (2023). Adeleke University Journal of Engineering and Technology (AUJET), Vol. 6, No. 1, pp. 177–187. ISSN 2714-2450.

A peer-reviewed engineering research paper presenting the design, development, and laboratory validation of an IoT-based gas monitoring and leakage detection system for LPG filling stations. Integrated an MQ-5 gas sensor, ATmega328-P microcontroller, SIM900L GSM module, and DHT11 temperature-humidity sensor with a cloud-connected web application built using Node.js, React, MongoDB, and the ThingSpeak API. Results confirmed accuracy and reliability under laboratory conditions, with future work proposed in machine learning integration for predictive gas explosion modelling.

Published Technical Writing

► [Finding Items in a Supermarket: A Story About Optimizing Geolocation in Software](#)

Adeyemi, A. W. (2025). Notion.

A technical analysis of when to delegate spatial computation to the database versus resolving it in application memory. Benchmarked against a real-world logistics platform, the pattern reduced task acceptance validation time by 95% (from ~100ms to under 5ms) by eliminating redundant database round-trips.

► [How to Run a PostgreSQL Database in Azure Kubernetes Service and Integrate It with a Node.js Express](#)

[Application](#)

Adeyemi, A. W. (2024). freeCodeCamp.

A walkthrough of containerised database deployment on Kubernetes, covering persistent volume configuration, service networking, environment variable management, and production-grade connection pooling for cloud-native database operations.

► [How to Build a QR Code Generator for URLs with Node.js, Next.js, and Azure Blob Storage](#)

Adeyemi, A. W. (2024). freeCodeCamp.

A technical guide on integrating server-side rendering, RESTful API design, and enterprise cloud storage into a reproducible architectural pattern for cloud-agnostic environments.

PROFESSIONAL EXPERIENCE

Backend Developer

Sep 2025 – Present

[Independent Purchasing Company](#) | Lagos, Nigeria (Full-Time)

Backend engineer on Relay Pilot, a large-scale logistics platform serving thousands of concurrent users across riders, restaurants, and delivery operators.

- Architected a modular backend system with 50+ domain modules using NestJS and clean architecture principles across logistics, payments, authentication, orders, and notifications.
- Led development of an enterprise real-time messaging platform with Socket.IO supporting 10,000+ concurrent users with sub-second latency, including multi-device support and distributed conversation management.
- Built a multi-provider payment processing system integrating Paystack and 9JaPay with webhook signature verification, virtual account management, and MongoDB ACID-compliant transactions — processing millions in financial transactions at 99.99% data consistency.
- Established a distributed queue system using BullMQ and Redis processing 10+ specialized job queues with exponential backoff retry logic and 99.9% uptime.
- Scaled real-time logistics tracking using Redis-backed Socket.IO with geospatial indexing for live rider location updates across 1000+ concurrent connections.

Backend Developer

May 2025 – Sep 2025

[Luftsocial](#) • Remote (Upwork)

Architected backend systems for an AI-driven social media management platform featuring intelligent content workflows and multi-channel notification infrastructure.

- **Integrated LangChain and OpenAI** with custom credit tracking, subscription-based usage limits, and real-time consumption monitoring — maintaining 99.8% service availability across AI-powered features.
- Architected scalable microservice-ready backend with 20+ domain modules using NestJS and Domain-Driven Design principles, maintaining 95% code modularity across subscription, AI, and platform integration domains.
- Integrated a multi-provider payment abstraction layer unifying Stripe and Paystack with a unified webhook processing pipeline, handling 15+ payment event types at 99.95% reliability.
- Built a real-time messaging system with Socket.IO supporting 10,000+ concurrent users featuring typed event handlers, room-based broadcasting, offline message sync, and multi-device support.
- Contributed to containerized deployment workflows using a Docker-based pipeline with multi-environment configurations, reducing deployment time by 75% across dev, staging, and production.

Backend Developer

Feb 2025 – Present

[Verify Your Customer](#) | Lagos, Nigeria (Contract)

Architecting a production-grade digital identity verification backend serving 10,000+ users.

- Implemented authentication infrastructure covering JWT tokens, role-based access control, and Google OAuth.
- Engineered TLS certificate pinning with public key verification and custom CA chain validation, eliminating man-in-the-middle attack vectors for critical webhook communications.
- Designed a comprehensive database abstraction layer with dynamic query building, reducing code duplication by 60% across 12+ data models while improving query performance.
- Architected event-driven webhook delivery using Bull Queue and Redis, processing 1000+ asynchronous jobs at 99.9% reliability through exponential backoff retry logic.

- Integrated Stripe Payment Intents API processing \$10K+ in monthly transactions with metadata tracking and automated subscription activation.

Backend Developer

Aug 2024 – May 2025

[Sokozuri Africa](#) • [Lagos, Nigeria](#) (Contract)

Backend services and database optimization for a multi-vendor e-commerce platform serving African artisans and entrepreneurs.

- Engineered a Redis caching layer with automatic response interception and 1-hour TTL, reducing API response times from 500ms to 150ms and database load by 60%.
- Managed 36+ database migrations with complex relational schemas supporting multi-vendor e-commerce operations, ensuring data integrity across 15+ interconnected entities.
- Integrated AWS S3 and DigitalOcean Spaces with secure pre-signed URL handling, supporting 30MB uploads and managing 10,000+ product images at scale.
- Designed scalable MVC architecture with layered separation of concerns, improving code maintainability and enabling parallel development.
- Built a Socket.io-based real-time messaging system with user presence tracking and online status management.

Backend Developer

Jun 2024 – Jan 2025

[Uwana Energy](#) | [Lagos, Nigeria](#)

Backend development for a clean-energy platform connecting renewable energy providers with consumers — directly supporting Nigeria's clean-energy transition through scalable software infrastructure.

- Architected a scalable Node.js REST API using clean layered architecture, organising 45+ controllers across domain-driven business modules and enabling parallel development across teams.
- Implemented multi-database architecture supporting MySQL, PostgreSQL, and SQLite using Sequelize ORM, enabling flexible deployment across development, staging, and production environments.
- Built an end-to-end GitLab CI/CD pipeline with Docker containerization and automated deployments to AWS ECS, reducing deployment time by 75%.
- Integrated Paystack payment gateway with webhook signature verification and dynamic fee calculation, processing \$50K+ in transactions at 99.9% uptime.
- Architected a scalable transactional email service processing 5,000+ daily emails at 98% deliverability.

Backend Developer

May 2021 – Jul 2022

[Qoinpal](#) | [Remote](#)

- Architected a multi-blockchain cryptocurrency service in Go, implementing ECDSA key pair generation, multi-stage hashing algorithms (SHA256, RIPEMD160, Keccak256), and address encoding for Bitcoin and Ethereum networks.
- Designed RESTful API endpoints with structured JSON responses and comprehensive error handling for wallet generation services.
- Integrated BadgerDB embedded database with transaction-based operations and data consistency guarantees, experience with embedded systems and low-level database management.

TEACHING & MENTORSHIP EXPERIENCE

Software Engineering Instructor & Senior Mentor

2023 – 2024

[Nupat Technologies](#) | [Lagos, Nigeria](#)

Lead instructor and mentor for incoming cohorts of software engineering interns, responsible for designing and delivering a structured onboarding curriculum and building a culture of technical accountability.

- Designed and delivered structured technical curricula covering codebase setup, database management, cloud environment configuration (AWS, Azure, DigitalOcean), DevOps pipeline implementation, and core software engineering principles.
- Conducted individual mentorship sessions to identify knowledge gaps, tailor learning approaches, and provide ongoing one-on-one guidance.
- Supervised project-based learning cycles, evaluated intern deliverables, and provided structured written and verbal feedback.
- Maintained instructional documentation including setup guides, troubleshooting playbooks, and annotated code examples, ensuring institutional knowledge was retained beyond any individual's tenure.

Academic Tutor & Laboratory Supervisor

2019 – 2022

[Kwara State University](#) | [Nigeria](#) | [Department of Electrical & Electronics Engineering](#)

Selected by faculty to lead weekly tutorial and laboratory sessions for undergraduate cohorts of 13–30 students across multiple course units.

- Facilitated structured weekly tutorial sessions, clarifying lecture content, working through problem sets, and addressing student misconceptions in real time.
- Supervised laboratory practicals, ensuring safe and productive use of equipment and guiding students through experimental procedures.
- Developed supplementary study materials and revision guides tailored to common points of difficulty, empowering students to resolve lower-level issues independently.

Research Assistant

2022

[Kwara State University](#) | [Nigeria](#)

- Supported faculty in supervising undergraduate final-year projects, reviewing methodologies and providing structured feedback on technical reports.
- Assisted in grading coursework and coordinating academic resources, maintaining confidentiality and upholding institutional assessment standards.
- Served as a point of contact for students navigating research processes, fielding questions and directing students to appropriate faculty resources.

CONFERENCES, WORKSHOPS & PROFESSIONAL DEVELOPMENT

- **Oscafest Open-Source Festival** - Participated in open-source software showcases, workshops, and networking sessions, gaining exposure to modern software development practices (2025)
- **SAIL Innovation Lab Start-up Accelerator Program** - Engaged in startup mentorship, product development sprints, and innovation workshops focused on tech entrepreneurship (2024)

- **Lagos State Tech Fest** - Attended panels and hands-on sessions on emerging technologies, including AI, cloud computing, and software engineering trends (2023)
- **Kwara State University Tech Fest** - Participated in student-led hackathons, tech talks, and workshops on programming, robotics, and software development (2022)

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- [AWS APAC Solutions Architecture](#) – Amazon Web Services (2024)
- [Azure Kubernetes Service](#) – KodeKloud (2024)
- [Quantum Software Engineering Virtual Experience](#) – Quantum (2024)
- [Cisco Software Engineering Virtual Experience](#) – Cisco (2022)
- [Accenture North America Virtual Coding Experience](#) – Forage / Accenture (2022)
- [Introduction to IoT and Digital Transformation](#) – Cisco (2022)

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Go, Bash

Backend Frameworks: NestJS, Express, FastAPI, Socket.IO, BullMQ

Databases & Storage: MySQL, PostgreSQL, MongoDB, Redis, MinIO, AWS S3; Sequelize, TypeORM, Prisma, Mongoose

AI & ML: LangChain, OpenAI API, Flowise AI

Cloud & DevOps: AWS (ECS, S3, Lambda), Azure (AKS, Blob Storage), Docker, Kubernetes, GitLab CI/CD, Digital Ocean

Architecture: Clean Architecture, Domain-Driven Design, Microservices, Event-Driven Systems, REST API Design

Tools: Postman, Git, VS Code, Notion